

HW-1, Due 10/4

1. A person pulls a sled (mass $m = 10 \text{ kg}$) at constant velocity along a horizontal surface with a rope at angle $\theta = 45^\circ$ above horizontal. The coefficient of kinetic friction is $\mu = 0.2$ and $g = 10 \text{ m/s}^2$. How much work does the person do over a distance of 30 m?
2. A 10 kg object starts from rest. A force of 50 N is applied at 37° above the horizontal for 2.0 s. Given $g = 10 \text{ m/s}^2$, $\mu_k = 0.2$, find:
 - (a) The work done by the applied force.
 - (b) The work done by friction.
3. An object moves at speed v on a horizontal surface. After traveling distance S , its speed is $v/3$. The resistive force is constant. What total distance does it travel before stopping?
4. A 2.0 kg block is pushed horizontally by a constant 6.0 N force on a table with $\mu_k = 0.25$. How much does its kinetic energy increase after moving 5.0 m? Take $g = 10 \text{ m/s}^2$.
5. A uniform rod of length L and weight W lies flat on the ground. It is raised upright and placed on a table of height h . What minimum work is required?
6. A mass m falls freely. How much gravitational potential energy is lost during the t th second?

7. A camera is released from height 100 m while a balloon rises at 5.0 m/s. Neglect air resistance and take $g = 10 \text{ m/s}^2$. What is the impact speed?